Week 3: The Avellaneda-Stoikov Model

Optimal market making via stochastic control – derived from scratch

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Industry Mentor: Dendi Suhubdy (Bitwyre) | **Guest: Fenni Kang (Coincall)**

Contents

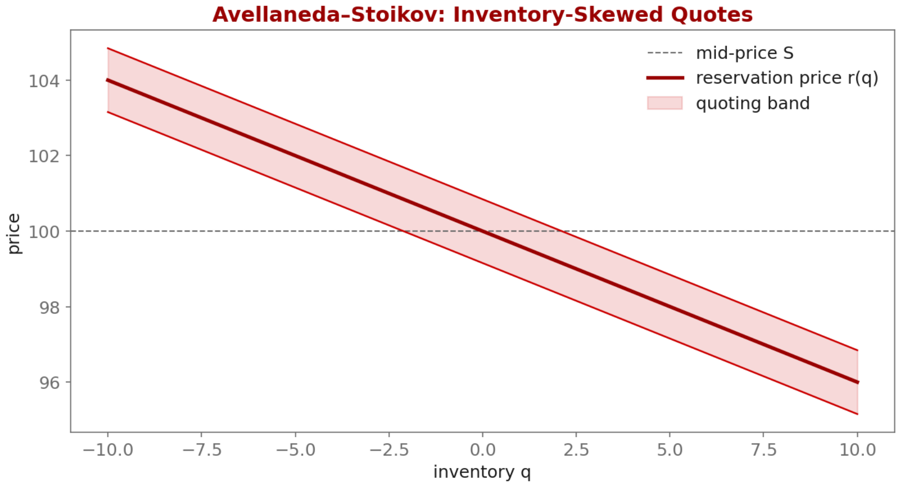
Learning Objectives

- Set up the Avellaneda-Stoikov control problem
- Follow the HJB derivation and understand every modeling choice
- Interpret the reservation price and the optimal half-spread

- Mid-price: dS_t = sigma dW_t (arithmetic Brownian motion)
- Fill intensity: lambda(delta) = A exp(-kappa delta)
- Wealth from fills; inventory q jumps by +/-1 on a fill
- Exponential (CARA) utility of terminal wealth

- Dynamic programming -> HJB partial differential equation
- Separation ansatz collapses HJB to ODEs in inventory
- Asymptotic (long-horizon) closed-form quotes
- Reservation price skews quotes against inventory

Illustration



Derivation & Rationale

$$dS_t = \sigma \, dW_t$$

Why: Arithmetic (not geometric) Brownian motion is chosen because over a short market-making horizon price changes are small and roughly additive; it keeps the HJB linear in S and makes the cash dynamics exact.

Derivation & Rationale

$$\lambda(\delta) = A\, e^{-\kappa\delta}$$

Why: Fill intensity decays with quote distance δ. The exponential form is the simplest law consistent with 'further from mid $\Rightarrow$ fewer fills'; A sets the base rate, κ the decay. It also yields a closed-form optimum.

Derivation & Rationale

$$\max\ \mathbb{E}\left[-e^{-\gamma\,(X_T + q_T\,S_T)}\right]$$

Why: CARA (exponential) utility is used because constant absolute risk aversion makes the value function factor out wealth, so the optimal quotes do not depend on the cash level – exactly what we want for a market maker.

Derivation & Rationale

$$\partial_t V + \frac{1}{2}\sigma^2 \partial_{SS} V + \max_{\delta^b} \lambda(\delta^b)\left[V(q{+}1, x{-}S{+}\delta^b) - V\right] + \max_{\delta^a} \lambda(\delta^a)\left[V(q{-}1, x{+}S{+}\delta^a) - V\right] = 0$$

Why: The HJB equation: the diffusion term prices inventory risk; each max term is the dealer optimally choosing the bid/ask offset, trading fill probability against captured spread.

Derivation & Rationale

$$V(t, x, S, q) = -e^{-\gamma(x+qS)}\,\theta(t, q)$$

Why: This ansatz is not a guess from nowhere: under CARA + arithmetic dynamics the value function inherits a multiplicative structure across (x, qS), so all the wealth/price dependence is exponential and only $\theta(t, q)$ remains to solve.

Derivation & Rationale

$$r(t, q) = S - q\gamma\sigma^2(T - t)$$

Why: The reservation (indifference) price: it sits BELOW mid when you are long ($q > 0$) so you quote to sell down inventory. The skew scales with risk aversion γ, variance σ^2, and time left – more risk $\Rightarrow$ more skew.

$$\delta^* = \frac{1}{\gamma}\ln\left(1 + \frac{\gamma}{\kappa}\right) + \frac{\gamma\sigma^2}{2}(T - t)$$

Why: The optimal half-spread = an order-flow term $\frac{1}{\gamma}\ln(1+\gamma/\kappa)$ (how hard fills are to get) plus an inventory-risk term $\frac{\gamma\sigma^2}{2}(T-t)$. You widen when risk-averse, when volatile, and when far from the horizon.

Getting the Data from Coincall

- This week uses simulated dynamics (no live data needed).
- For realistic sigma, estimate it from Coincall perp klines:
  - `GET /open/option/market/kline/history/v1/{name}?period=m1 (or futures klines)`
- You will calibrate A, kappa from real fills in Week 5.

Cross-check exact paths at https://docs.coincall.com/

```
import torch

def as_quotes(S, q, gamma, sigma, kappa, tau):
    """Asymptotic Avellaneda-Stoikov bid/ask. Vectorized over inventory q."""
    reservation = S - q * gamma * sigma**2 * tau
    half_spread = (1.0/gamma) * torch.log1p(gamma/kappa) \
                  + 0.5 * gamma * sigma**2 * tau
    return reservation - half_spread, reservation + half_spread # bid, ask

S = torch.tensor(100.0); q = torch.arange(-10, 11).float()
bid, ask = as_quotes(S, q, gamma=0.1, sigma=2.0, kappa=1.5, tau=1.0)
print("inventory ->", q.tolist())
print("bid ->", bid.round(decimals=2).tolist())
```

Summary — Key Takeaways

- Model: arithmetic mid, exponential fill intensity, CARA utility
- HJB + the exponential ansatz collapse the problem to ODEs in inventory
- Reservation price skews quotes against inventory; r = S - q*gamma*sigma^2*(T-t)
- Half-spread = order-flow term + inventory-risk term
- Every modeling choice is made for tractability and has a clear interpretation

Glossary — Key Terms

Reservation price	inventory-adjusted indifference price that quotes are centered on
Half-spread	distance from the reservation price to each quote
HJB equation	the PDE characterizing the optimal value function
Ansatz	an assumed functional form that reduces the HJB to ODEs
CARA utility	constant absolute risk aversion, U(w) = -exp(-gamma w)
Intensity lambda(delta)	Poisson fill rate as a function of quote distance delta
Risk aversion gamma	parameter setting inventory skew and spread width

Reading

- Avellaneda-Stoikov (2008); Cartea-Jaikumar-Penalva, Ch. 10

References

- Avellaneda, M., & Stoikov, S. (2008). High-frequency trading in a limit order book. Quantitative Finance, 8(3), 217-224.
- Gueant, O. (2016). The Financial Mathematics of Market Liquidity, Ch. 4. Chapman & Hall/CRC.
- Cartea, Jaikumar, & Penalva (2015). Algorithmic and High-Frequency Trading, Ch. 10.